

Diagnosis and management of Guillian Barre Syndrome in children and adolescents

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Executive summary -

The diagnosis of GBS should be based on clinical characteristics, a relevant constellation of -1
.clinical findings, and the exclusion of other relevant potential causes

:Requirements for the diagnosis of sensory- motor or motor GBS are

Bilateral flaccid paralysis of extremities •

Absent or decreased deep tendon reflexes in affected limbs •

Progressive worsening for no more than 4 weeks **(strong recommendation)** •

Diagnostic lumbar puncture with subsequent CSF protein analysis and a cell count is done **when -2**
the diagnosis is uncertain or when an alternative diagnosis needs to be excluded. Results
supportive of GBS are an increased CSF protein concentration, and a normal or only slightly
increased CSF white blood cell count (usually <5 cells/ μ L, rarely 5–50 cells/ μ L (known as cyto-
albuminological dissociation) . A CSF white blood cell count of >50 cells/ μ L should raise suspicion
for alternative diagnoses **(strong recommendation)**

We suggest that for diagnostic certainty in the event of an initially-normal CSF finding, a lumbar
puncture may be repeated 7-10 days after the first symptoms appear, if it is deemed necessary due
to unclear clinical or electrophysiological results. (Good practice statement)

Electrophysiological examination for confirming GBS diagnosis is recommended as it is -3
necessary for obtaining the highest level of diagnostic certainty (Strong recommendation). **If the**
.diagnostic tool is not available, refer the patient to an appropriate clinical centre

We suggest that for confirmation of diagnosis in the event of an initially normal electrophysiological
finding, the procedure may be repeated 1 or 2 weeks later **if it is deemed necessary because of**
other unclear data. (conditional recommendation)

Performing MRI of brain and spinal cord is done when the available findings are ambiguous or if -4
there is evidence of pathology at either the spinal level (e.g., bladder dysfunction at the onset of
disease, definable motor or sensory lesion) or central level (encephalopathy, pyramidal signs)
(strong recommendation)

Microbiological and serological diagnostics may be carried out as the diagnosis of GBS is more -5
likely if there is a history of recent infection. Asking about these antecedent events may increase
diagnostic confidence and is generally more useful when present than absent (Conditional
recommendation)



It is not advised to do routine testing for serum antibodies against gangliosides in most patients -6 with motor-sensory GBS, because of their low-moderate sensitivity and frequent delay in reporting results of antibody assays beyond the therapeutic window (good practice statement)

If MFS (or MFS spectrum) is suspected, test for serum antibodies against GQ1b can be done especially when there is some clinical doubt and test results can be obtained within reasonable time (Conditional Recommendation)

Starting IVIg or plasma exchange as soon as possible is recommended in patients unable to walk -7 unaided (GBS- DS grade 3 or more) if still within the first 2 weeks from onset of weakness (strong recommendation)

Using the most frequently used and proven effective standard course of IVIg (0.4 g/kg/day for 5 • days) rather than a low- dose (0.4 g/kg/day for 3 days) or a high- dose (0.4 g/kg/day for 6 days) regimen or a 2- day regimen (1 g/kg/day) or using four to five plasma exchanges over 1– 2 weeks with a total exchanged volume of 12– 15 L (good practice statement)

We advise to start IVIg (or PE four to five exchanges) in patients who are still able to walk unaided (GBS- DS grade 2) within 4 weeks from onset of weakness, **but who have a fast rate of deterioration, a risk of requiring ventilatory support, swallowing difficulties, autonomic disturbances or poor prognostic factors** (good practice statement)

We don't recommend the use of oral corticosteroids or IV methylprednisone (IVMP) for the -9 treatment of GBS. (strong recommendation)

We don't recommend using eculizumab for treatment of GBS.(conditional recommendation) -10 because of the lack of demonstrated efficacy, the known adverse effects (all patients had some .adverse effects) and the high cost currently

We advise using a prognostic model at hospital admission to quantify the risk of requiring -11 mechanical ventilation. This can be quantified using the modified Erasmus GBS Respiratory Insufficiency Score (mEGRIS)(table4) OR

:Using the following risk factors for requiring mechanical ventilation during hospital admission

Rapid progression of limb weakness during hospital admission

GBS- DS grade 4 (unable to walk 5 m even with aid)

Neck flexion, facial, or bulbar weakness

The inability to cough

Autonomic instability such as fluctuations in blood pressure or heart rate. (good practice statement)



We advise regularly assessing any decline in respiratory function by measuring forced vital capacity (FVC), and single breath count (SBC). Fig (good practice statement)

Changing the diagnosis from GBS to A-CIDP is suggested after a few weeks from onset in some -12 patients initially diagnosed with GBS, especially if the patient worsens after initial improvement or stabilization (known as a treatment-related fluctuation, TRF), or presents as mild or slowly progressive GBS and continues to worsen.(conditional recommendation)

.A-CIDP is more likely if there are three (or more) TRFs

.A-CIDP cannot be confirmed unless there is further worsening at least 8 weeks after onset

In case of a TRF, consider re-treatment with IVIg or PE.(good practice statement)



Recommendations -

?PICO 1: What are the diagnostic criteria of GBS

We recommend that diagnosis of GBS should be based on clinical characteristics, a relevant constellation of clinical findings, and the exclusion of other relevant potential causes

:Requirements for the diagnosis of sensory- motor or motor GBS are

Bilateral flaccid paralysis of extremities •

Absent or decreased deep tendon reflexes in affected limbs •

Progressive worsening for no more than 4 weeks (strong recommendation, high grade evidence) •

:Evidence Summary

:Features that support diagnosis

Relative symmetry, 2-Relatively mild/absent sensory symptoms and signs,3- Cranial nerve-1 involvement (especially bilateral facial palsy), 4- Autonomic dysfunction, 5- Respiratory insufficiency (due to muscle weakness), 6- Pain (muscular/radicular in back or limb),7- Recent history of infection (<6 weeks).⁽¹⁶⁾

To ensure the highest level of diagnostic certainty, additional proof is required in the form of increased protein levels in the CSF, along with a mildly increased cell count, and .electrophysiological signs of neuropathy

The maximal duration of progression originates from large studies showing that progression does not exceed 2 weeks in most patients, and almost never exceeds 4 weeks.(9,10)

:Cerebrospinal fluid analysis -2

PICO2: Cerebrospinal fluid (CSF) examination: In patients with clinically suspected GBS, does examination of the CSF compared with no CSF examination influence the diagnostic accuracy, ?treatment response and patient outcome

We strongly recommend a diagnostic lumbar puncture with subsequent CSF protein analysis -1 and a cell count **particularly when the diagnosis is uncertain or when an alternative diagnosis needs to be excluded** (strong recommendation, moderate grade evidence)

Results supportive of GBS are an increased CSF protein concentration, and a normal or only -2 slightly increased CSF white blood cell count usually <5 cells/ μ L, rarely 5–50 cells/ μ L (known as cyto-albuminous dissociation) (Strong recommendation, moderate grade evidence)



A CSF white blood cell count of >50 cells/ μ L should raise suspicion for alternative diagnoses -3
(Strong recommendation, moderate grade evidence)

We suggest that for diagnostic certainty in the event of an initially-normal CSF finding, a lumbar -4
puncture may be repeated 7-10 days after the first symptoms appear, **if it is deemed necessary due
to unclear clinical or electrophysiological results**. (Good practice statement)

:Evidence summary

Normal CSF protein does not exclude GBS. It should be noted that protein concentration is often -
normal during the first week, and only increases with disease progression so, diagnostic sensitivity
of an increased CSF protein depends on the time CSF is examined after onset of weakness. After 7-
10 days, it generally reaches a level above that of the reference value, which notably differs in
.children according to age [11]

The specificity of a raised CSF protein is unknown, and it does not rule out some mimics of GBS -
as CSF protein levels can be elevated in other disorders that are similar to GBS.. Therefore, a
normal CSF protein reading does not rule out GBS, while an elevated protein level is not solely
sufficient to confirm diagnosis. [12]

Nevertheless, the CSF cell count is usually normal, although occasionally it can be mildly elevated -
to <50/mm³, rarely more [13]. In such cases, other differential diagnoses as meningitis should be
considered as possible causes [13]. Due to the fact that the GBS-related increase in protein is
dependent on time, and the standard CSF value for protein depends on age, the cell count can
.potentially be more valuable as a differential diagnostic factor than protein level

Both CSF protein and cell count may be artefactually raised after IVIG [14] -

Since this is an invasive procedure that can cause discomfort and be stressful for children and -
often has no therapeutic consequences, it is preferred not to be repeated except if unclear clinical
or electrophysiological results. [15]

:Electrophysiological examination -3

*PICO 3: In patients with clinically suspected GBS, Regarding the added value of electrophysiological
?examination*

We recommend performing electrophysiological examination for confirming GBS diagnosis as -1
it is necessary for obtaining the highest level of diagnostic certainty (strong recommendation, low
grade evidence)

..If the diagnostic tool is not available, refer the patient to an appropriate clinical centre

We suggest that for confirmation of diagnosis in the event of an initially normal -2
electrophysiological finding, the procedure may be repeated 1 or 2 weeks later **if it is deemed
necessary because of other unclear data**. (conditional recommendation, moderate grade
evidence)



:Evidence summary

Prospective and retrospective studies that evaluated up to 84 patients with clinically suspected GBS or up to 66 AIDP cases with a variable number of controls concluded that numerous nerve conduction parameters are abnormal early after disease onset.(19,20) However, some studies included small numbers of patients and/or did not include controls. Sensitivity and specificity vary .according to electrodiagnostic criteria and the control groups used

A typical electrodiagnostic finding is required in order to reach the highest level of diagnostic certainty described by Sejvar et al. [18] Such examinations can be painful and are often difficult to perform in children. Moreover, the necessary expertise is often lacking in pediatric hospitals. Nonetheless, nerve conduction studies and, to a lesser extent, EMG, are required for distinguishing between GBS variants. Notably, due to the latent appearance of pathological spontaneous activity .it only makes sense to perform an EMG 2-3 weeks after onset

Although electrophysiological criteria for the diagnosis of GBS and its variants have been repeatedly defined, they are not always consistently applied and cannot be considered definitive .[21,22]

MRI brain and spine -4

PICO (4) In patients with clinically suspected GBS, does using MRI brain and/or spine compared . with not using MRI studies influence the diagnostic accuracy

We recommend performing MRI of the brain and spinal cord when the available findings are ambiguous or if there is evidence of pathology at either the spinal level (e.g., bladder dysfunction at the onset of disease, definable motor or sensory lesion) or central level (encephalopathy, pyramidal signs). (strong recommendation, low grade evidence)

:Evidence summary

Although a spinal MRI is primarily carried out to exclude space-occupying lesions or any other pathology, a contrast enhanced MRI procedure could also support the diagnosis of GBS. A number of publications has reported pathological uptake of contrast agent, particularly by the ventral nerve roots and cranial nerves, hence suggesting that a contrast-enhanced MRI could serve as a diagnostic criterium for GBS in the face of persistently unremarkable CSF and nerve conduction findings.(23,24)

In a retrospective study by Yikilmaz et al. 38 of 40 patients with childhood GBS showed contrast enhancement of the nerve roots; Mulkey et al. reported 22 of 24 patients with spinal nerve root enhancement in an initial MRI(25,26). However, increased uptake of contrast agent is also possible in other inflammatory conditions (serous meningitis, neuroborreliosis) and neurometabolic disorders (Krabbe disease). Since prospective examinations have not yet been carried out, the sensitivity and specificity of contrast-enhanced MRI remains unclear in children and adolescents .with GBS



: Antecedent events -5

PICO (5) In patients with clinically suspected GBS, does enquiring about antecedent events (infection or vaccination) compared with not enquiring about antecedent events influence the ?diagnostic test accuracy, treatment response and patient outcome

:Infection

We suggest that microbiological and serological diagnostics may be carried out as the diagnosis of GBS is more likely if there is a history of recent infection. Asking about these antecedent events may increase diagnostic confidence and is generally more useful when present than absent (Conditional recommendation, moderate grade evidence)

:Evidence Summary

About two- thirds of patients who develop GBS report symptoms of an infection in the 6 weeks preceding the onset of the condition ⁽²⁷⁾ These infections are thought to trigger the immune response that causes GBS. In childhood GBS, a preceding infection is identified as the potential .causative agent in 60-70% of cases

Six pathogens have been temporally associated with GBS in case-control studies: Campylobacter jejuni, cytomegalovirus, hepatitis E virus, Mycoplasma pneumoniae, Epstein-Barr virus and Zika virus.^(28,29) It has been suggested that other pathogens are linked to GBS on the basis of evidence .from case series or epidemiological studies, but their role in the pathogenesis of GBS is uncertain

GBS is not or marginally increased in incidence after SARS- CoV- 2 infections although large case-control studies are lacking, from the epidemiological and cohort studies that exist.^(30,31)

In general, the absence of an antecedent illness does not exclude a diagnosis of GBS, as putative .infections or other immunological stimuli can be subclinical

: Vaccination

We recommend thorough microbiological and serological diagnostic tests if GBS occurs within 1-6 weeks of vaccination for investigating a potential link with the vaccinating antigen, as well as with alternative causal agents. (Strong recommendation, moderate grade evidence)

:Evidence Summary

The results arising from microbiological and serological diagnostic tests can have substantial legal .implications at the social and health care levels

GBS has been associated temporally with numerous vaccines; however, such temporal association must be differentiated from causality. In general, specific biological markers indicative of a cause-and-effect association with a particular pathogen or vaccine are absent.⁽³²⁾



Well-conducted case-control studies have been able to show that most vaccines have a very unlikely causal link to GBS. The only exception comes from a swine flu vaccine that was used in an American immunization campaign in 1976-77 and led to a significant increase in the number of GBS cases. This effect was not observed again in later immunization campaigns, even after vaccination during the recent swine flu pandemic in 2009-10.⁽³³⁾

Furthermore, an accumulation of GBS cases was observed 30-40 years ago following immunization with a rabies vaccine that was grown on mammalian brain tissue. This, however, has not been observed in association with modern rabies vaccines.⁽³⁴⁾

Serum antibodies-6

PICO (6) In patients with clinically suspected GBS, does testing for serum antibodies compared with not testing for these antibodies influence the diagnostic test accuracy, particularly for GBS ?subtypes such as AMAN and Miller Fisher syndrome

We don't advise to do routine testing for serum antibodies against gangliosides in most patients with motor-sensory GBS, because of their low-moderate sensitivity and frequent delay in reporting results of antibody assays beyond the therapeutic window (good practice statement)

We suggest testing for serum antibodies against GQ1b in patients suspected to have MFS (or MFS spectrum) especially, when there is some clinical doubt and test results can be obtained within reasonable time (Conditional Recommendation, moderate grade evidence)

:Summary of evidence

Serum antibodies that have been tested in GBS cases include GM1, GM1b, GD1a, GQ1b, GalNAc-GD1a, GT1a, GD1b, GD3, O-GD3, GT3, O-GT3, sulfatide, galactocerebroside, CNTN1, NF155 and NF186, cardiolipin, LM1, sulphated glycolipids, P0 and PMP22. The test accuracy varies depending on GBS subtype, tested antigen and control group. ^(35,36)

In GBS, the sensitivity of anti-ganglioside antibody panels is reported to vary between 32% and 64%, depending on the presence of a recent infection or GBS subtype (AIDP or AMAN).⁽³⁷⁾

For MFS, sensitivity for anti-GQ1b antibodies is high (88%–100%), with a very high specificity (100%). Especially, when there is some clinical doubt and test results can be obtained within reasonable time, testing for anti-GQ1b antibodies is considered helpful. ^(38,39)

:Therapeutic recommendations -1

PICO (7) In patients with clinically suspected GBS ,Is treatment with intravenous immunoglobulins ?(IVIG) recommended in children with GBS

We recommend starting IVIg as soon as possible in patients unable to walk unaided (GBS- DS grade 3 or more) if still within the first 2 weeks from onset of weakness (strong recommendation- high level of evidence)



Using the most frequently used and proven effective standard course of IVIg (0.4 g/kg/day for 5 • days) rather than a low- dose (0.4 g/kg/day for 3 days) or a high- dose (0.4 g/kg/day for 6 days) regimen or a 2- day regimen (1 g/kg/day) (good practice statement)

We advise to start IVIg (or PE) also in patients who are still able to walk unaided (GBS- DS grade 2) within 4 weeks from onset of weakness, **but who have a fast rate of deterioration, a risk of requiring ventilatory support, swallowing difficulties, autonomic disturbances or poor prognostic factors** (good practice statement)

: Summary of evidence

In the absence of systematic dose-ranging study data, IVIG is usually administered in children as it is in adults with acute GBS, namely with a single cycle of 2 g/kg body weight distributed over 4 - 5 consecutive days. Reducing the period of IVIG administration to 2 g/kg in 2 days did not show any improved effects but was associated with a higher frequency of relapses ^[40]

The most common side-effects were headache (12.8% of infusions), fatigue (5.2%), stomach-ache (2.3%) and myalgia (2.3%). Acute side-effects such as skin rash, fever, bronchospasm and thoracic pain occurred more rarely (0.3-0.6%) ^[41]

An initially progressive disease course with a subsequent plateau phase without recovery is classified as severe GBS, for which the optimal treatment approach remains unclear . However, in children a satisfactory recovery can still be expected even after a prolonged plateau-phase of up to 90 days^[42]

Plasmapheresis, plasma exchange-8

?PICO (8) Is treatment with plasmapheresis recommended in children with GBS

The same indications as listed in the IVIG in cases when IVIG is not available. We recommend four to five exchanges over 1– 2 weeks with a total exchanged volume of 12– 15 L in patients who are severely disabled (unable to walk unaided, bedridden or ventilated). (strong recommendation, high grade evidence)

We suggest PE (four to five exchanges over 1– 2 weeks) in patients who are still ambulatory, but who have a fast rate of deterioration, a risk of requiring ventilatory support, swallowing difficulties or other poor prognostic factors. These patients are considered at high risk of further deterioration, which may potentially be prevented by starting treatment early(good practice statement)

Summary of evidence

The best effect of PE is seen when therapy is started within seven days of symptom onset, although .an effect can still be detected if applied up to 30 days post-onset

The given dosages for PE mostly comprise 4-5 cycles with an exchange volume of 200-250 ml/kg .body weight over 7-14 days



·Albumin is a more favorable exchange fluid than FFP ^[43]

There is no preference for treatment with either IVIg or PE. Therefore, the choice may depend on the availability or the experience.⁽⁵⁰⁾ However, IVIg is generally associated with few adverse events and is readily available in most hospitals. PE requires special facilities, good intravenous access and
·has a slightly higher adverse event (AE) rate ^[42]

The combination of PE followed by IVIG did not yield any significant additional benefits, although
·an effect in individual patients cannot be ruled out

·Application of PE within 1-2 weeks of the preceding IVIG therapy is not indicated

PICO (9). In patients diagnosed with GBS, does treatment with corticosteroids (IV or oral), alone or ? in combination with IVIg or PE, influence the disease course

We don't recommend the use of oral corticosteroids or IV methylprednisone (IVMP) for the treatment of GBS.(strong recommendation, moderate level of evidence)

:Summary of evidence

There is a probable lack of efficacy of IVMP (moderate certainty evidence from a trial with 242 patients),⁽⁴⁴⁾

The probable lack of efficacy of the combination of IVIg and IVMP compared with IVIg and placebo (moderate certainty evidence from a trial with 225 patients)⁽⁴⁵⁾

The probable harm (delayed recovery) of oral corticosteroids (low certainty evidence from 4 trials with a total of 120 patients) using various oral regimens of the equivalent of 40 mg prednisolone daily for at least 2 weeks, and the high certainty evidence of adverse effects (diabetes more common), despite that hypertension was less common in the corticosteroid-treated patients, which is crucial in decision making. ^(46,47,48)

PICO (10). Does treatment with pharmacological therapies other than PE, IVIg (as eculizumab, ? mycophenolate, interferons or other) influence disease course

We don't recommend the use of immunosuppressive treatment approaches as an alternative to immunomodulation with plasmapheresis and IVIG. (conditional recommendation, low grade evidence)

:Summary of evidence

As there is evidence for complement activation from pathological studies and from animal models of GBS.⁽⁵³⁾ Eculizumab, a complement blocking agent, was beneficial in an animal model.⁽⁵⁴⁾

In two small phase 2 trials (including a total of 41 patients), no beneficial effects of eculizumab could be demonstrated (very low to low certainty evidence).⁽⁵⁵⁾



PICO (11) In patients diagnosed with GBS, what is indication of ICU admission and mechanical ventilation?

We advise using a prognostic model at hospital admission to quantify the risk of requiring mechanical ventilation. This can be quantified using the modified Erasmus GBS Respiratory Insufficiency Score (mEGRIS)(table5) OR

Using the following risk factors for requiring mechanical ventilation during hospital admission. :These include

Rapid progression of limb weakness during hospital admission -

GBS- DS grade 4 (unable to walk 5 m even with aid) -

Neck flexion, facial or bulbar weakness -

The inability to cough -

Autonomic instability such as fluctuations in blood pressure or heart rate. (good practice - statement)

We advise regularly assessing any decline in respiratory function by measuring forced vital capacity (FVC), and single breath count (SBC). (good practice statement) Fig 2

Assessment and monitoring of GBS

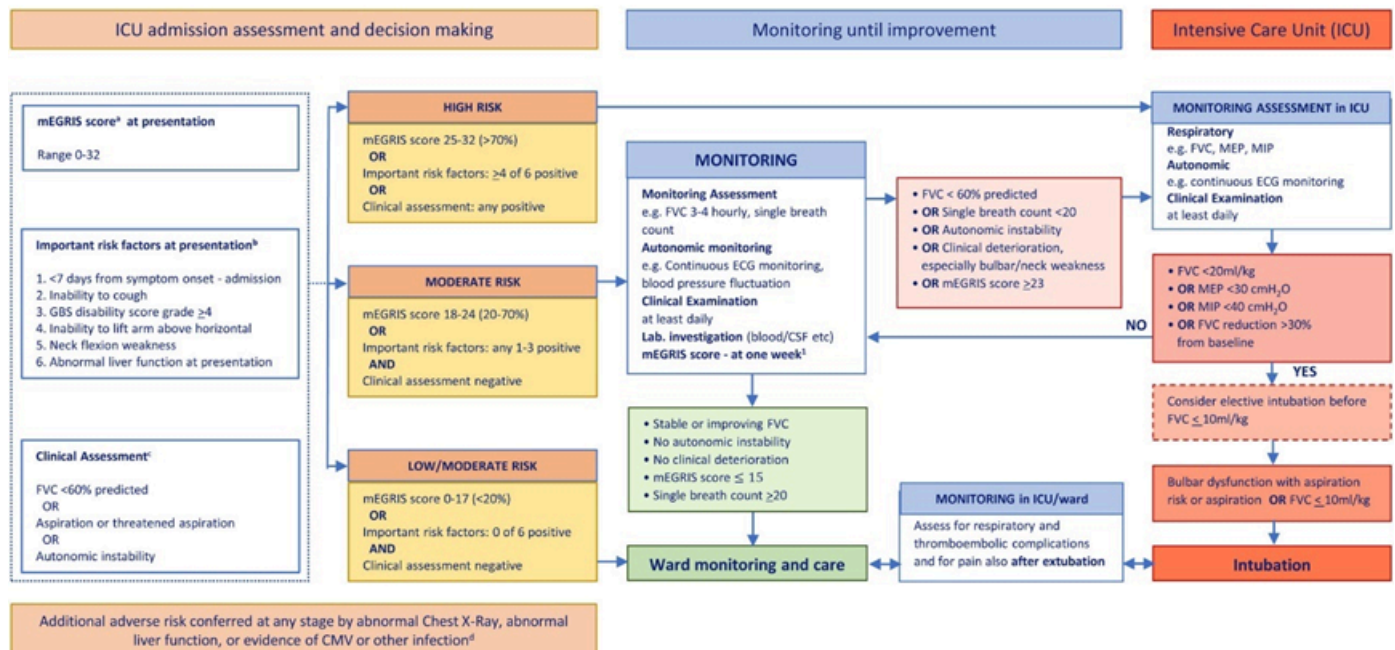


Fig.2: Assessment and monitoring of GBS

:Frequent monitoring of the following quantitative measures of ventilatory function

a- FVC should be checked between three and six times a day, depending on severity, and until significant worsening seems unlikely. While the patient is still declining and there is a reduction in .FVC, 4- hourly monitoring is likely appropriate



- b- A fall of the FVC > 30% below the predicted baseline should alert concern, a fall of >30% in 24 h likely indicates immediate transfer to ICU, or a 50% decline in under 24 h likely indicates the need .for ventilation
- c- Single breath count <20 (inability to count in a single breath out loud from 1 to 20 indicates the .need to transfer to ICU

: Summary of evidence

According to the published case series on GBS in childhood, 15-25% of patients were dependent on mechanical ventilation. Respiratory failure in children particularly needs to be reckoned with when the disease progresses quickly and the upper extremities and cranial nerves are affected ^(56,57) Moreover following discharge from the ICU the possibility of secondary deterioration in the .ensuing weeks should be kept in mind

PICO (12). In patients initially diagnosed with GBS, does the presence of certain clinical symptoms or laboratory features compared with their absence predict the subsequent diagnosis of A-CIDP as ?confirmed by neurological worsening at >8 weeks from onset

We suggest changing the diagnosis from GBS to A-CIDP after a few weeks from onset in some patients initially diagnosed with GBS, especially if the patient worsens after initial improvement or stabilization (known as a treatment-related fluctuation, TRF), or presents as mild or slowly progressive GBS and continues to worsen.(conditional recommendation, low grade evidence)

.A-CIDP is more likely if there are three (or more) TRFs •

.A-CIDP cannot be confirmed unless there is further worsening at least 8 weeks after onset •

In case of a TRF, consider re-treatment with IVIg or PE.(good practice statement) •

:Evidence summary

In a prospective study of 170 patients initially diagnosed as GBS, 16 (9%) had a TRF and another 8 (5%) were subsequently diagnosed with A-CIDP ⁽⁵⁸⁾. Confirmation of these results however warrants .additional studies in larger numbers of patients

A-CIDP is likely in patients initially diagnosed with GBS, if there is further worsening after 8 weeks from onset (sensitivity 100%, specificity 92%)^(51,52), or when there are three (or more) TRFs (episodes of worsening following treatment-induced improvement/stabilisation) (sensitivity 52%, .specificity 96%)^(49,59)

Reduced MNCV < 90% of lower limit of normal (or <85% if small distal CMAP) was more frequently found in A-CIDP than in GBS . Patients with A-CIDP tested with nerve US within 4 weeks of onset had greater enlargement of peripheral nerves compared with those with GBS (sensitivity 88%, specificity 84%)⁽⁶⁰⁾. Some patients suspected to have A-CIDP may have an autoimmune nodopathy ⁽⁶¹⁾. These patients have a poorer response to conventional therapies for CIDP, and there is



anecdotal evidence that these patients may response to rituximab. There is no evidence from an RCT, but observational data indicate that a repeated course of IVIg or PE can be effective in case of a TRF⁽⁶²⁾

Rationale: It is important to diagnose A-CIDP because treatment differs from GBS. A-CIDP however should not be overdiagnosed in severely weak patients with slow or no improvement. If in doubt, the presence of muscle wasting and denervation on electromyography indicates that GBS-related .axonal degeneration is more likely than A-CIDP



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Abbreviations -

AE: Adverse events

AIDP :acute inflammatory demyelinating polyneuropathy

. AMAN :acute motor axonal neuropathy

. AMSAN :acute motor and sensory axonal neuropathy

BBE: Bickerstaff brainstem encephalitis

CIDP: Chronic inflammatory demyelinating polyneuropathy

CSF :cerebrospinal fluid

DS: Disability Scale

FFP: Fresh Frozen Plasma

FVC :forced vital capacity

GBS :Guillain Barre syndrome

.ICU :intensive care unit

IVIG :intravenous immunoglobulins

IVMP : IV methylprednisolone

mEGRIS :modified Erasmus GBS Respiratory Insufficiency Score

MFS :Miller Fisher Syndrome

MRC: Medical Research Council

PE :plasmapheresis

SBC: single breath count

,TRF :treatment-related fluctuation



Introduction -

Guillain Barre syndrome (GBS) is the most common cause of acute flaccid paralysis with an annual global incidence of approximately

per 100 000 person years(1,2) The diagnosis of GBS relies upon a combination of clinical 2–1 features, often with support of electrodiagnostic and laboratory features. Most diagnostic criteria for GBS require a combination of history, neurological examination, cerebrospinal fluid (CSF)and electrodiagnostic results.(3)

GBS can be subclassified into a number of variants, depending on clinical presentation and electrophysiological findings (4) .The classic variants are acute demyelinating inflammatory .polyneuropathy (AIDP) and acute motor axonal neuropathy (AMAN)

These are characterized by rapidly-progressing, ascending symmetrical weakness, with attenuation or loss of muscle proprioceptive reflexes. Among pediatric patients in the peak phase of the disease, 75% can no longer walk unaided, 30% are tetraparetic,35-50% show cranial nerve involvement, and 15-20% have respiratory failure and/or autonomic dysfunction. Furthermore, up to 70% suffer from neuropathic pain, which can be severe and occasionally occur as the first symptom. Localized variants of GBS include Miller Fisher Syndrome (MFS) (cranial nerve affection and ataxia, areflexia, serum anti-GQ1b antibody detection in >90% cases) and the pharyngeal-cervical-brachial variant (predominantly bulbar and neck weakness, serum IgG antibodies against GT1a frequently detected), both of which are extremely rare in childhood (5) .Immunomodulatory therapy with plasmapheresis (PE) or intravenous immunoglobulins (IVIG) has been proven effective in GBS namely by accelerating the improvement of weakness. (6,7)

The improvement follows a plateau phase that ranges from days to weeks and runs an extremely variable course, regardless of whether a specific therapy has been applied or not. The overall long-term prognosis for children with GBS is more favorable than that in adults, whereby the majority of children largely regain motor function.(8)

There are several forms of GBS within the disease spectrum, which are defined by the clinical features, with some overlap: fig 1

Motor-sensory((weakness with sensory symptoms and/or signs) and motor ((weakness without sensory symptoms or signs)GBS are considered 'typical' GBS -1

Pharyngeal–cervical–brachial GBS with weakness in the corresponding regions, bilateral facial weakness with limb paraesthesias (but no limb weakness) -2

Paraparetic variant with weakness starting in the legs that may evolve to tetraparesis -3



Pure sensory syndrome of numbness or tingling, reduced or absent reflexes, and sometimes ataxia or pseudoathetosis, but no weakness, with a time course like GBS. Nerve conduction in these patients may show signs of demyelination in sensory or motor nerves. Although the patients with this pure sensory syndrome do not fulfil all diagnostic criteria for GBS, this probably represents a rare sensory variant of GBS

MFS spectrum : Patients with typical MFS have ophthalmoplegia, ataxia and areflexia

Bickerstaff brainstem encephalitis (BBE) is considered a rare variant of MFS that clinically has a combination of ophthalmoplegia, ataxia, pyramidal tract signs and impaired consciousness

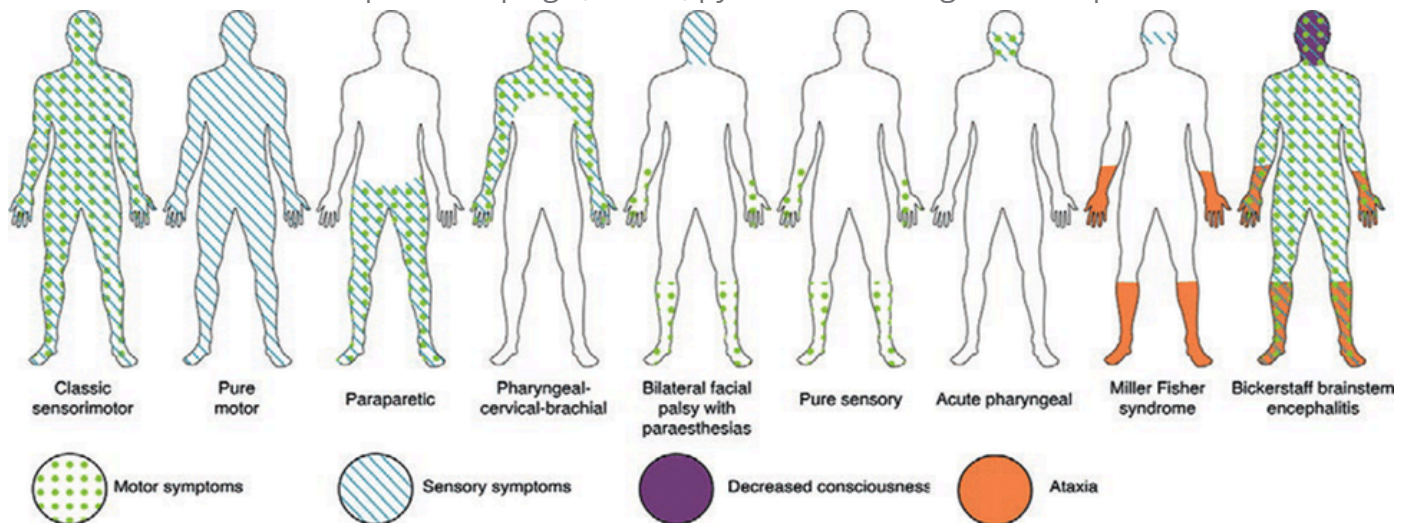


Fig 1: Forms of GBS



Scope and purpose -

The aim of this guideline is to provide practical recommendations for diagnostic procedures and the treatment of GBS in childhood and adolescence. The target population are children and adolescents presenting with clinical features suggestive of GBS. The aim is to optimize diagnostic .accuracy and to improve the patient outcomes



Target audience -

This guideline is intended for pediatricians, intensivists, pediatric neurologists and general .practitioners who are involved in the care of pediatric population



Evidence assessment -

According to WHO Handbook for Guidelines, we used the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) approach to assess the quality of a body of evidence, develop and report recommendations. GRADE methods are used by WHO because these represent internationally agreed standards for making transparent recommendations. Detailed GRADE :information is available on the following sites

[/GRADE working group: https://www.gradeworkinggroup.org](https://www.gradeworkinggroup.org) •

[/GRADE online training modules: http://cebgrade.mcmaster.ca](http://cebgrade.mcmaster.ca) •

:Table 1 Quality and Significance of the four levels of evidence in GRADE

Quality	Definition	Implications
High	The guideline development group is very confident that the true effect lies close to that of the estimate of the effect	Further research is very unlikely to change confidence in the estimate of effect
Moderate	The guideline development group is moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different	Further research is likely to have an important impact on confidence in the estimate of effect and may change the estimate
Low	Confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the true effect	Further research is very likely to have an important impact on confidence in the estimate of effect and is unlikely to change the estimate
Very low	The group has very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of the effect	Any estimate of effect is very uncertain

Table 2 Factors that determine How to upgrade or downgrade the quality of evidence



Downgrade in presence of	Upgrade in presence of
Study limitations -1 Serious limitations -2 Very serious limitations	Dose-response gradient +1 Evidence of a dose-response gradient
Consistency -1 Important inconsistency	Direction of plausible bias +1 All plausible confounders would have reduced the effect
Directness -1 Some uncertainty -2 Major uncertainty	Magnitude of the effect +1 Strong, no plausible confounders, consistent and direct evidence
Precision -1 Imprecise data	+2 Very strong, no major threats to validity and direct evidence
Reporting bias -1 High probability of reporting bias	

The strength of the recommendation

The strength of a recommendation communicates the importance of adherence to the recommendation.

Strong recommendations

With strong recommendations, the guideline communicates the message that the desirable effects of adherence to the recommendation outweigh the undesirable effects. This means that in most situations the recommendation can be adopted as policy.

Conditional recommendations

These are made when there is greater uncertainty about the four factors above or if local adaptation has to account for a greater variety in values and preferences, or when resource use makes the intervention suitable for some, but not for other locations. This means that there is a need for substantial debate and involvement of stakeholders before this recommendation can be adopted as policy.

When not to make recommendations

When there is lack of evidence on the effectiveness of an intervention, it may be appropriate not to make a recommendation.



Methods -

This guideline has been developed by a group of dedicated pediatric neurology consultants volunteering from all over the Egyptian universities across Egypt. It is the result of an initiative held by the Egyptian Health Council, aiming to formulate evidence-based recommendations for clinical practice.

A comprehensive search for guidelines was undertaken to identify the most relevant guidelines to consider for adaptation.

Inclusion/ exclusion criteria followed in the search and retrieval of guidelines to be adapted

- Selecting only evidence-based guidelines (guideline must include a report on systematic literature searches and explicit links between individual recommendations and their supporting evidence)

- Selecting only national and/or international guidelines

- Specific range of dates for publication (using Guidelines published or updated in 2015 and later)

- Selecting peer reviewed publications only

- Selecting guidelines written in English language

- Excluding guidelines written by a single author, not on behalf of an organization to be valid and comprehensive, a guideline ideally requires multidisciplinary input

- Excluding guidelines published without references as the panel needs to know whether a thorough literature review was conducted and whether current evidence was used in the preparation of the recommendations

The following characteristics of the retrieved guidelines were summarized in a table

- Developing organization/authors

- Date of publication, posting, and release

- Country/language of publication

- Date of posting and/or release

- Dates of the search used by the source guideline developers



All retrieved Guidelines were screened and appraised using AGREE II instrument (www.agreetrust.org) by at least three members. The panel decided on a cut-off point or ranked the guidelines (any guideline scoring above 50% on the rigor dimension was retained)

The Guideline Development Group (GDG) decided to adapt the Guideline published by the German Speaking Society of Neuropediatrics (GNP) "Diagnosis and Treatment of Guillain Barre Syndrome in Childhood and adolescence: An evidence and Consensus Based guideline- 2020 **AND** the guideline published by the European Academy of Neurology/Peripheral Nerve Society on the Diagnosis and treatment of Guillain Barre syndrome- 2023.(16,17)



Clinical indicators for monitoring -

- :Documentation of symptoms and signs in the medical file-1
- ..Progressive weakness of arms and legs for no more than 4 weeks
- .By examination: absent or decreased deep tendon reflexes in affected limbs
- .Electrophysiological studies (showing evidence of neuropathy) if available -2
- CSF analysis (showing cyto-albuminuous dissociation), if diagnosis is uncertain -3
- IVIg (0.4 g/kg/day for 5 days) or plasma exchange 4-5 sessions should be started if GBS-DS grade 3 or more in first 2 weeks OR grade 2 in patients who have a fast rate of deterioration, a risk of requiring ventilatory support, swallowing difficulties, autonomic disturbances or poor prognostic factors -4
- Documentation of the risk factors for requiring mechanical ventilation -5
- Rapid progression of limb weakness during hospital admission-*
 - GBS- DS grade 4 (unable to walk 5 m even with aid) -*
 - Neck flexion, facial or bulbar weakness -*
 - The inability to cough -*
 - .Autonomic instability such as fluctuations in blood pressure or heart rate -*
- .Single breath count (SBC) to assess any decline in respiratory function-6



Research Gaps -

:Recommended further research in the field of GBS

In Pathogenesis and Immune Mechanisms .1

Better understanding of molecular mimicry between pathogens (e.g., *Campylobacter jejuni*) and .gangliosides

.Role of complement activation, T-cell subsets, and innate immunity in different GBS variants

.Exploration of genetic susceptibility (e.g., HLA subtypes, complement-regulatory genes)

.Role of infections and vaccines

Biomarkers and Early Diagnosis .2

Development of CSF and blood biomarkers

.Neurofilament light chain (NfL) and anti-ganglioside antibodies as prognostic tools

.Advanced imaging (e.g., nerve ultrasound, DTI MRI) to detect early inflammation or axonal injury

Treatment options .3

:Trials of novel immunotherapies, such as

Complement inhibitors (e.g., eculizumab)

Anti-GM1 monoclonal antibodies

T-cell modulation strategies



Guideline update -

.This guideline will be updated whenever there is new evidence



Appendix -

Table3 :GBS disability scale (Hughes and Cornblath).(45)

healthy - 0
minor symptoms or signs of neuropathy, but capable of manual work and running - 1
can walk without the aid of a stick for 5 m across an open space, but is not capable of manual work or running - 2
can walk with a stick, orthosis or support (5 m across an open space) - 3
bedridden or wheelchair-bound - 4
ventilation assistance required (for any part of the day or night) - 5
dead - 6

Table 4:Medical Research Council (MRC) Scale for Manual Muscle Testing

patient can maintain position against maximal resistance and through the entire physiological range of motion of the joint - 5
patient can maintain position against moderate resistance, and moves actively through the entire physiological range of motion of the joint - 4
patient cannot maintain position against resistance, but can move the extremity against gravity through the full range of motion - 3
patient can move the extremity through part of the physiological range of motion if gravity is eliminated - 2
muscle contraction can be detected by palpation if gravity is eliminated - 1
no contractions identifiable - 0

Table 5. The erasmus Guillain–Barré syndrome (GBS) Respiratory Insufficiency Score (eGRIS)

Score	Categories	Measures
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0	7>	Days between onset of weakness and hospital admission
1	4-7	
2	3>	
0	Absent	Facial and/or bulbar weakness at hospital admission
1	Present	
0	60-51	MRC sum score at hospital admission
1	50-41	
2	40-31	
3	30-21	
4	20>	
0-7	NA	EGRIS

.NA, not applicable

An eGRIS of 0–2 indicates a low risk of mechanical intervention, 3–4 indicates an intermediate risk .of mechanical intervention and ≥5 indicates a high risk of mechanical intervention

Table 6: The most important differential diagnoses of GBS

Intracranial

- Meningeosis neoplastica/leucaemica
- Brain stem encephalitis

Peripheral Nerves

- Axonal sub-acute recurrent neuropathy, with elevated CSF lactate levels in patients with a PDHc1a-mutation
- Metabolic disorders such as hypermagnesemia or hypophosphatemia
- Tick paralysis
- Heavy metal toxicity such as arsenic, gold and thallium
- Medication-induced neuropathy (e.g. vincristine, platinum compounds, nitrofurantoin, paclitaxel)
- Porphyria
- Critical illness neuropathy
- Vasculitis



Diphtheria ·

Spinal cord

Infarction, myelitis, compression ·

Anterior horn motor neurons Polio and other enteroviruses that can trigger poliomyelitis, ·
including West Nile Virus

Nerve roots

Chronic inflammatory demyelinating polyneuropathy (CIDP) ·

Cauda equina compression ·

Neuromuscular end-plate

Myasthenia gravis ·

Organophosphate poisoning ·

Botulism ·

Muscle

Critical illness ·

Polymyositis ·

Dermatomyositis ·

Hypo-/hyperkalemia ·

Periodic paralysis ·



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